

PT SATYAMITRA SURYA PERKASA Engineering Procurement & Construction		PROCEDURE, ITP, HIRADC PROJECT 5&6 TANJUNG JATI 'B' EXPANSION (JAWA-4) COAL FIRED STEAM POWER PLANT 2X1000 MW								
No.	TRANSMITTAL CONTENT	NO DOCUMENT	KODE	REV	DATE APPROVED	STATUS	TRANSMITTAL NUMBER	DATE SEND	ITEM	REMARKS
25	SHORING PROTECTION CALCULATION FOR SOIL EXCAVATION AT RCP Ø 1500 mm AREA	SSPPM-MS/7016	PNMS-016	0		Not yet replied	SSPTJB5&6TR-ENG0093	19-Mar-17	PM-MS	



Project : TANJUNG JATI-B EXP. (JAWA 4) CFSPP 2 x 1000 MW

Location : TUBANAN, JEPARA - CENTRAL JAVA

TRANSMITTAL SHEETS

TO : MES - MURINDA JOINT OPERATION
ATTN : Mr.PATRICK W SOULE
FROM : SSP
CC : Ir. Sulaiman S

DATE : May 19th, 2017

NO : SSP/TJB5&6/TR-ENG/0093

JOB NO : 1703.C

Please find the enclosed : ☐ Letter(s) ☐ Drawing (s) ☐ Sample(s)

☒ Document(s) ☐ Calculation(s) ☐ Other(s)

Which are transmitted for you :

☐ File ☐ Information ☐ Reference ☒ Approval ☐ Comments

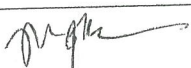
Responded by :		Comments :	
Sent by :		Panca Widyatmoko. ST	May 19th, 2017
Received by :			
		Name	Signature
			Date

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SHORING PROTECTION CALCULATION FOR SOIL EXCAVATION AT RCP Ø1500mm AREA

DOCUMENT NUMBER :
SSP/PM-MS/17/016

Prepared by
 Muh Dwi R.

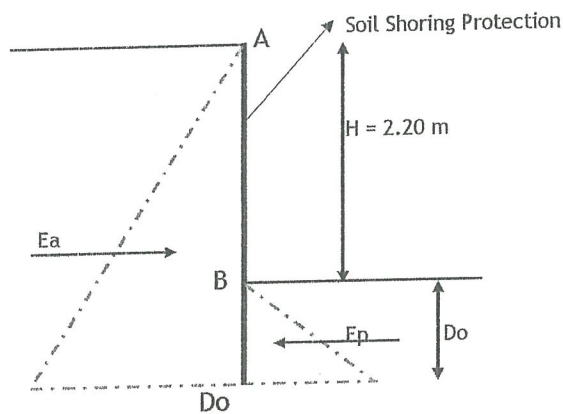
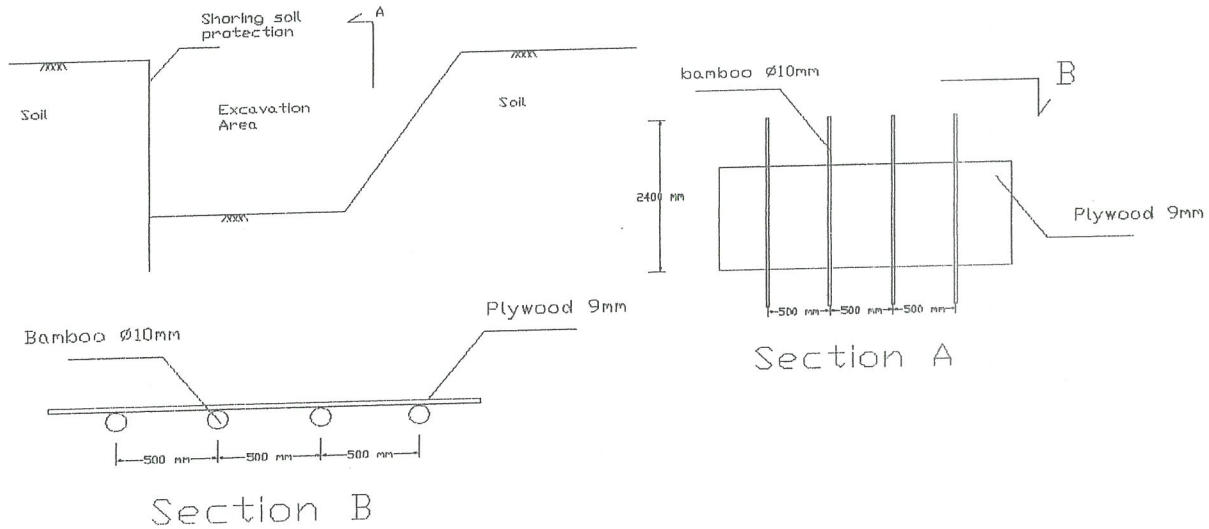
Rev. No.	Date	Description	Approved by MESJO
0	May 19, 2017	Issued for approval	

CLIENT	: MES - JOINT OPERATION		
PROJECT	: TANJUNG JATI B EXPANSION (JAWA-4) 2x1000 MW - SITE DEVELOPMENT	PROJECT NO. :	1703.C
WORK/ SCOPE	: GENERAL CIVIL WORKS		
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Calculation Shoring Protection For Soil Excation Area RCP Ø1500



Soil Properties

$$\gamma = 1.5 \text{ T/m}^3 = 15 \text{ KN/m}^3$$

$$\theta = 40^\circ$$

Shoring Properties

Plywood	t = 9mm	Span of bamboo stud (S1)	= 50 cm
Allow Bending Stress (wood)	FW = 100 kg/cm ²	Allow Bending Stress (bamboo FB)	= 760 kg/cm ²
Modulus elasticity (wood)	EW = 100000 kg/cm ²	Bamboo size	= Ø10 cm

Calculation For 1m length

$$k_a = \tan^2(45 - (\theta/2))$$

$$k_a = \tan^2(45 - (40/2)) = 0.217$$

$$k_a = \tan^2(45 + (\theta/2))$$

$$k_a = \tan^2(45 + (40/2)) = 4.6$$

$$E_a = 0.5 \times \gamma \times k_a \times (H + D_o)^2$$

$$e_a = 1/3 \times (H + D_o)^2$$

$$E_p = 0.5 \times \gamma \times k_a \times D_o^2$$

$$e_p = 1/3 \times D_o^2$$



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$$\Sigma MD = 0$$

$$E_a \cdot e_a - E_p \cdot e_p = 0$$

$$(0,5 \times 16 \times 0,217 \times (2,2 + D_o)^2) \times (1/3 \times (2,2 + D_o)^2) - (0,5 \times 16 \times 4,16 \times D_o^2) \times (1/3 \times D_o^2)$$

$$\Rightarrow D_o^3 - 0,36 D_o^2 - 0,81 D_o - 0,59 = 0$$

Trial and error, $D_o = 1,34$ m

So, the depth of bamboo inside soil = 1.34 m

Total length of bamboo = $2,20 + 1,34 = 3,54$ m

M act Max at distance "x" from point B

$$\Sigma M_{total} = -9,86x^3 + 3,58x^2 + 7,93x + 5,78 = 0$$

Deferensial to find kuadrat equation

$$(d\Sigma Mx / dx) = 0$$

$$\Rightarrow -29,58x^2 + 7,16x + 7,93 = 0$$

abc formula to get "x" value

$$x = 0,66 \text{ m from point B}$$

subs to the equation

$$\Sigma M_{total} = -9,86x^3 + 3,58x^2 + 7,93x + 5,78 = 0$$

$$= 9,74 \text{ KNm}$$

for 50 cm length of shoring

$$= 9,74 / 2$$

$$= 4,87 \text{ KNm}$$

Check for the Shoring Plywood 9mm and bamboo $\varnothing 10$ cm

Check plywood

$$\begin{aligned} \text{Momen inertia plywood} \quad I &= (1/12) bh^3 \\ &= (1/12) \times 50 \times 0,9^3 \\ &= 3,04 \text{ cm}^4 \\ c &= 0,9/2 \\ &= 0,45 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Momen allowable} \quad M_{allow} &= (FW \times I) / c \\ &= (100 \times 3,04) / 0,45 \\ &= 675,56 \text{ kgcm} \end{aligned}$$

Check the bamboo $\varnothing 10$ cm

$$\begin{aligned} \text{Momen inertia bamboo} \quad I &= (3,14 / 64) \times (10^4 - 8^4) \\ &= 289,665 \text{ cm}^4 \\ c &= 10/2 \\ &= 5 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Momen allowable} \quad M_{allow} &= (FB \times I) / c \\ &= (760 \times 289,665) / 5 \\ &= 44029,08 \text{ kgcm} \end{aligned}$$

$$\text{Total } M_{allow} = 675,56 + 44029,08$$

$$= 44704,64 \text{ kgcm}$$

$$= 4,47 \text{ KNm}$$

$$< M_{act} = 4,87 \text{ KNm}$$

.....not safe!!

Because the shoring is not safe so need additional bamboo at the panel. It's should add 1 more bamboo

$$\text{Add 1 more bamboo} = 44029,08 \times 2$$

$$= 88058,16 \text{ Kgcm}$$

$$M_{total} = 88058,16 + 675,56$$

$$= 88733,72 \text{ kgcm}$$

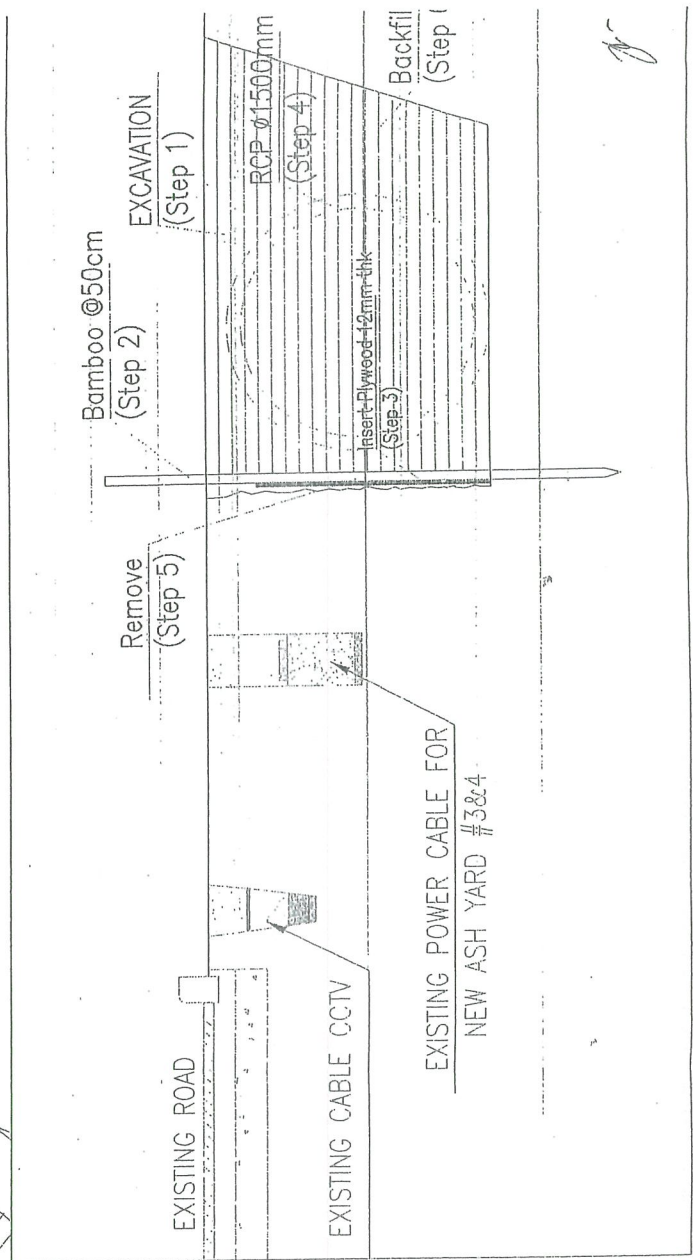
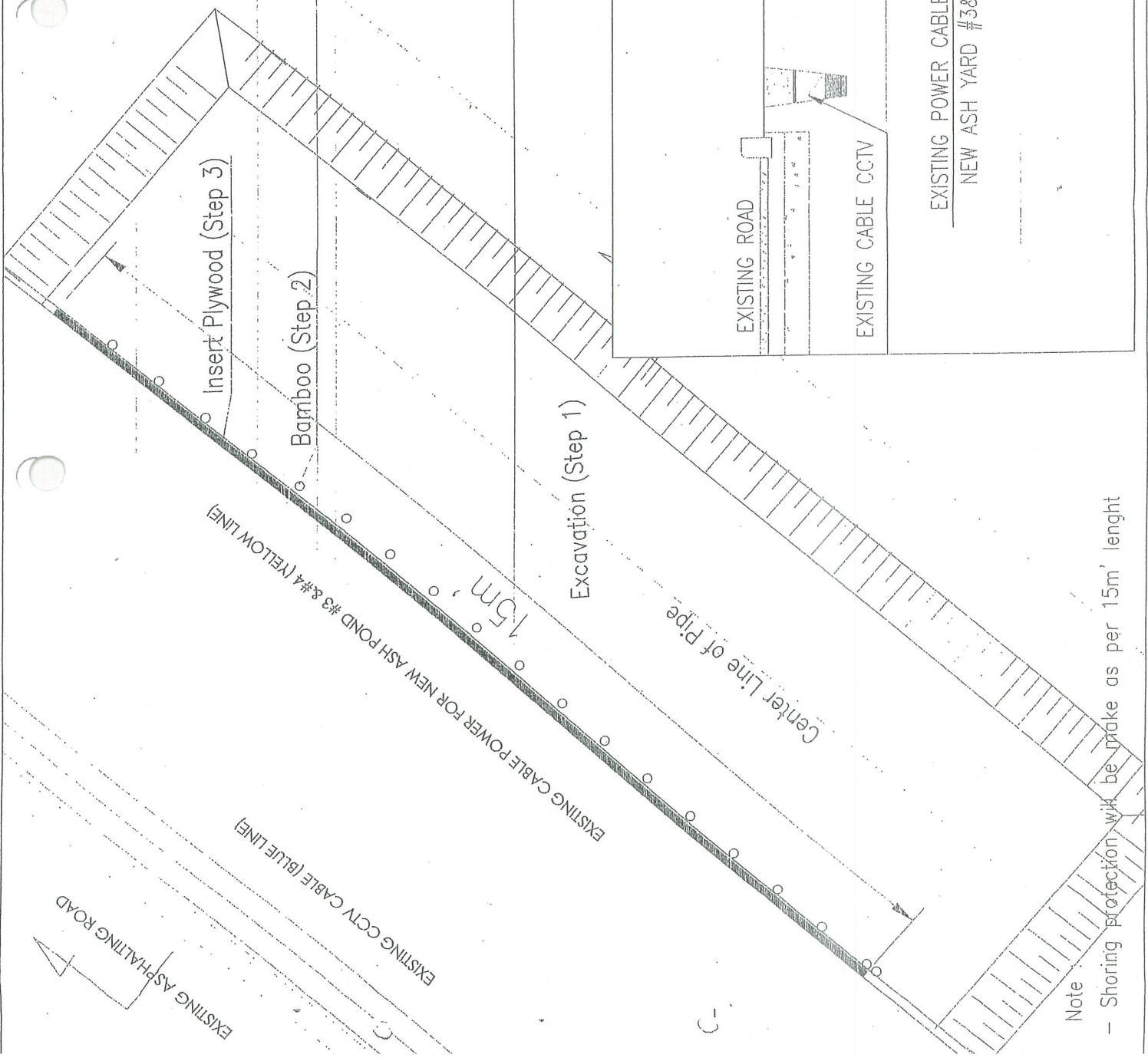
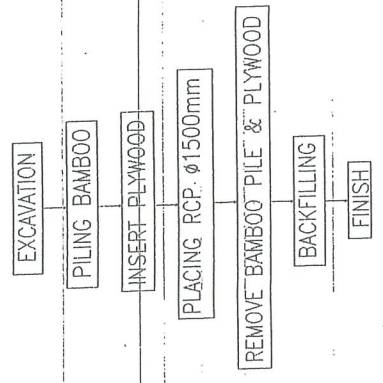
$$= 8,87 \text{ KNm}$$

$$> M_{act} = 4,87 \text{ KNm}$$

....safe

So 1 panel shoring protection need plywood 9mm and 2 bamboo $\varnothing 10$ cm every 50cm

FLOW CHART PIPE INSTALLATION AT CLOSE THE EXISTING CABLE



Note

- Shoring protection will be make as per 15m' lenght



THE WOOD DATABASE

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Bamboo



Common Name(s): Bamboo

Scientific Name: Hundreds of species among dozens of genera from the Poaceae (grass) family (Many timber-producing bamboos are from the Phyllostachys and Bambusa genera)

Distribution: Most timber-producing bamboos are from south Asia

Tree Size: Some of the largest bamboos can be up to 50-100 ft (15-30 m) tall, with a 3-6 in (10-20 cm) diameter

Average Dried Weight: 31 lbs/ft³ (500 kg/m³) to 53 lbs/ft³ (850 kg/m³)

Specific Gravity (Basic, 12% MC): .38 to .64, .50 to .85

Janka Hardness: 1,410 lb_f (6,270 N) to 1,610 lb_f (7,170 N)

Modulus of Rupture: 11,020 lb_f/in² (76.0 MPa) to 24,450 lb_f/in² (168.6 MPa)

Elastic Modulus: 2,610,000 lb_f/in² (18.00 GPa) to 2,900,000 lb_f/in² (20.00 GPa)

Crushing Strength: 8,990 lb_f/in² (62.0 MPa) to 13,490 lb_f/in² (93.0 MPa)

Shrinkage: Diameter: 10-16%, Wall Thickness: 15-17%

Bamboo (species unknown)

 [View more images below](#)

Color/Appearance: Generally a uniform and pale yellow to almost white. Live bamboo that has been left standing too long frequently develops fungal decay, discoloring the wood with brown or black streaks and patches.

Grain/Texture: Being a monocot in the grass family, bamboo does not have any sapwood/heartwood or growth rings. Texture is very uniform, and ranges from medium to fine depending on density. Bamboo that has been split and processed into lumber will have intermittent variations in the fiber at each node on the stem.

Endgrain: Resembles the endgrain of palm—another monocot.